TOPIC 31/60

Tokens vs. Words

The Lexical Building Blocks.

Why AI models don't read language the same way humans do, and how it impacts cost, latency, and reasoning.

Ashish Jangra
linkedin.com/in/ashish-jangra

THE ILLUSION

AI Models Don't Read Words

Humans process language via letters forming words. Neural networks cannot process text; they require structured mathematical inputs (numbers).

Enter the Tokenizer

Before text reaches an LLM, a "Tokenizer" chops a sentence into sub-word pieces called **Tokens** and assigns each piece an integer ID.

Input String:

"Hello world!"

Token Array (IDs):

[9906, 1917, 0]

Ashish Jangra
linkedin.com/in/ashish-jangra

THE METRIC

The 75% Rule

A token is generally equivalent to **~4 characters** of English text.

Common Words = 1 Token

Words that appear frequently in training data get their own dedicated integer.

"apple" → [1 token]

Complex Words = N Tokens

Rare or composite words get split into morphological chunks.

"hamburger" → [3 tokens]
ham + burg + er

100 Tokens ≈ 75 English Words

Ashish Jangra
linkedin.com/in/ashish-jangra

THE ALGORITHM

Byte-Pair Encoding (BPE)

How does the model decide where to split? Modern LLMs use a data compression technique called BPE.

// How BPE Works

BPE starts by treating every character as a token. It then scans billions of gigabytes of text and merges the most frequently adjacent pairs iteratively.

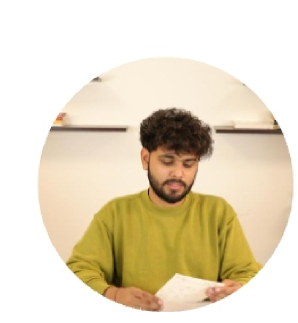
Step 1: 'e' and 'r' are merged into 'er'

Step 2: 't' and 'h' are merged into 'th'

Step N: 'th' and 'e' are merged into 'the'

Result: Common prefixes ('un-'), suffixes ('-ing'), and roots are optimized mathematically.

Ashish Jangra
linkedin.com/in/ashish-jangra



THE GLITCH

The Wordle Problem

Why do highly intelligent models like GPT-4 often fail at simple tasks like counting the number of "R"s in "Strawberry"?

Lexical Blindness

The model literally **cannot see the letters.** To the model, the word "Strawberry" is instantly compressed into a single integer ID (e.g., [49221]).

It knows that [49221] represents a red fruit, but unless it was explicitly trained on the spelling breakdown of that specific ID, it doesn't intuitively know that the ID contains three "r"s.

Ashish Jangra
linkedin.com/in/ashish-jangra

GLOBAL INEQUITY

The Multilingual Tax

Tokenizers are trained heavily on English internet data. This results in a massive inefficiency bias against non-English languages.

Cost to say "Hello World":

🇬🇧 English: Hello World	2 Tokens
🇪🇸 Spanish: Hola Mundo	3 Tokens
🇰🇷 Korean: 안녕 세상	6 Tokens

* Result: Using LLMs in Asian or marginalized languages costs significantly more API money and consumes context windows 3x faster.

Ashish Jangra
linkedin.com/in/ashish-jangra

SYNTAX TOKENIZATION

Whitespace is Expensive

Tokens don't just capture text; they capture spacing, punctuation, and indentation. This makes software code highly volatile for token consumption.

Standard Code:

```
if (x == 1) {
  return true;
}
```

Cost: ~12 Tokens

Minified Code:

```
if(x==1){return true;}
```

Cost: ~7 Tokens

* LLM Code generation utilizes massive amounts of tokens strictly to output spacebar formatting and tabs.

Ashish Jangra
linkedin.com/in/ashish-jangra

THE BUSINESS LOGIC

Token Economics

In the AI industry, the "Word" does not exist. Tokens are the fundamental currency of capability and cost.

1

API Billing

OpenAI, Anthropic, and Google bill strictly per 1,000 or 1,000,000 tokens. Not by word count.

2

Context Limits

A "128k Context Window" limits the model's memory to 128,000 tokens (approx 96,000 words), restricting how many documents you can upload.

Ashish Jangra
linkedin.com/in/ashish-jangra

VISUAL OUTPUT

How the AI sees it

A breakdown of how a complex sentence is fractured into sub-word tokens before being mapped to integers.

Token | ization | is | un | predict | able | !

Notice how "predictable" is sliced into "un", "predict", and "able". The model understands the semantic meaning of these root prefixes independently.

Ashish Jangra
linkedin.com/in/ashish-jangra

Tokens vs Words

CONCEPT UNLOCKED

🔢 Tokens are numerical IDs representing text chunks.

📏 1 Token roughly equals 0.75 English words.

💸 Non-English text drastically increases token counts & costs.

🧩 LLMs fail at spelling because they don't see the letters.

CONNECT WITH ME

LinkedIn

GitHub

Kaggle

Instagram

Ashish Jangra
linkedin.com/in/ashish-jangra